

Practice Quiz: Systems

Part A: Multiple Choice

1. The concept of a Markov Blanket in Active Inference primarily formalizes: A) The physical membrane of a cell B) The statistical boundary separating internal from external states C) The computational limits of an agent's processor D) The spatial extent of an organism's body
2. Autopoiesis, as defined by Maturana and Varela, refers to: A) The ability of a system to reproduce sexually B) A self-producing network that generates and maintains its own boundary C) The tendency of all systems to maximize entropy D) The process by which organisms evolve through natural selection
3. The life-mind continuity thesis claims that: A) Only organisms with nervous systems have minds B) Consciousness requires language and higher-order thought C) There is no sharp boundary between being alive and having a mind D) Machines will inevitably become conscious
4. In the Markov Blanket formalism, **sensory states** are: A) Internal states that encode the agent's beliefs B) States that mediate the influence of external states on internal states C) States that the agent uses to change the environment D) External states that the agent cannot access
5. The "fallacy of equivocation" objection to the FEP argues that: A) Free energy is not a real physical quantity B) The formal description of self-organization is being conflated with genuine cognition C) Markov Blankets cannot exist in continuous systems D) The FEP is unfalsifiable because it predicts everything
6. Clark and Chalmers' Extended Mind Hypothesis suggests that: A) The mind is confined to neural activity in the brain B) Cognitive processes can extend beyond the brain to include external tools and artifacts C) All physical objects have rudimentary mental properties D) The mind exists only as emergent social phenomena

7. From a Markov Blanket perspective, the Ship of Theseus paradox is best resolved by:
- A) Insisting that identity requires material continuity
 - B) Accepting that there are now two equally valid Ships of Theseus
 - C) Recognizing identity as constituted by dynamical pattern maintenance, not material substance
 - D) Rejecting the concept of identity altogether as philosophically incoherent

Part B: Short Answer

1. Explain how the concept of "surprise" in Active Inference differs from its everyday usage, and why minimizing surprise is essential for a system's persistence.
2. A candle flame maintains a stable form by consuming wax and oxygen. Does it have a Markov Blanket? Does it "resist dissolution"? What are the philosophical implications of your answer for the life-mind continuity thesis?
3. Construct a two-step argument: first, show how the Markov Blanket formalism captures the philosophical intuition behind autopoiesis; second, identify one aspect of autopoiesis that it *fails* to capture.